

Galaxy

Open, web-based platform for accessible, reproducible, and transparent
computational research

galaxyproject.org

Galaxy at a glance

400,000+

registered users

750,000

jobs per month

10,000+

analysis tools

22,000+

citations

>\$2M/year

free compute

3 continents

public servers

From genomics to climate science

- ▶ Genomics
- ▶ Transcriptomics
- ▶ Proteomics
- ▶ Metabolomics
- ▶ Single Cell
- ▶ Variant Analysis
- ▶ Genome Assembly
- ▶ Epigenetics
- ▶ Metagenomics
- ▶ Microbiome
- ▶ Machine Learning
- ▶ Imaging
- ▶ Climate Science
- ▶ Ecology
- ▶ Evolution
- ▶ Comp. Chemistry
- ▶ Materials Science
- ▶ Synthetic Biology
- ▶ FAIR Data
- ▶ One Health

No programming required

Galaxy provides a **point-and-click web interface** for complex analyses:

1. **Upload data** – from your computer, SRA, or the web
2. **Run a tool** – select from thousands of analysis tools
3. **Run a workflow** – use community-curated, tested workflows
4. **Interpret results** – integrated Jupyter, RStudio, and visualization tools

Every analysis is **automatically tracked** – full provenance and reproducibility built in.

A massive tool ecosystem

Galaxy integrates tightly with the open-source bioinformatics community:

Resource	Count
Galaxy tool wrappers	10,676
BioConda packages	11,699
BioContainers	12,333
CondaForge packages	30,722

All tools are free, open-source, and containerized for reproducibility.

Truly global infrastructure

The **usegalaxy.*** consortium provides free public servers across three continents:

- ▶ **usegalaxy.org** – United States (ACCESS-CC powered)
- ▶ **usegalaxy.eu** – Europe (de.NBI / ELIXIR)
- ▶ **usegalaxy.org.au** – Australia (Australian BioCommons)

Every user gets **250 GB permanent storage**, 1 TB scratch space, and 8 concurrent job slots – at no cost.

Galaxy Training Network (GTN)

The world's largest open bioinformatics training resource:

- ▶ **489 tutorials** across 35 topics
- ▶ **214 training videos**
- ▶ **499 contributors** worldwide
- ▶ Peer-reviewed, community-driven
- ▶ Galaxy Training Academy 2025:
3,700+ simultaneous participants
- ▶ Annual Galaxy Community Conference
- ▶ Free and open to all

Part of major international initiatives

AnVIL

NHGRI's Analysis, Visualization, and Informatics Lab-Space for genomic data science

BRC Analytics

Bioinformatics Resource Center for pathogen, host, and vector genomics

CFDE

NIH Common Fund Data Ecosystem – cloud workspace for protected datasets

VGP

Vertebrate Genomes Project – 600 chromosome-level assemblies and counting

VGP: a Galaxy success story

The Vertebrate Genomes Project uses Galaxy to produce near error-free, chromosome-level genomes:

- ▶ **600 assemblies** completed on Galaxy / Jetstream2
- ▶ Goal: reference genomes for all **~70,000** known vertebrate species
- ▶ Fully reproducible workflows available to the community

“95% of the main discoveries that have driven biotechnology came from studying things that were not model organisms at the time.”

– Giulio Formenti, Rockefeller University

The Galaxy ecosystem

Component	Role
IUC	Community-driven tool curation
IWC	Curated, tested workflow publishing
BioContainers	100,000+ reproducibility containers
Planemo	Tool development SDK
Pulsar	Distributed job execution (HPC to cloud)
TPV	Dynamic, intelligent job routing

Learn more

- ▶ galaxyproject.org
- ▶ usegalaxy.org
- ▶ training.galaxyproject.org
- ▶ github.com/galaxyproject

- ▶ usegalaxy.eu
- ▶ usegalaxy.org.au
- ▶ brc-analytics.org
- ▶ iwc.galaxyproject.org